1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: The clinical efficacy of intravesical BCG in high-grade non-muscle invasive bladder cancer (NMIBC) depends on the in situ formation of TLS-like lymphoid aggregates within the bladder mucosa after TURBT. TLS formation is more likely when residual tumor burden is low, enabling sufficient antigen deposition and local cytokine/chemokine signaling to recruit and organize B and T cells, dendritic cells, and follicular helper cells into ectopic lymphoid structures. These TLS-like aggregates support effective antigen presentation and targeted cytotoxic responses, yielding durable local control and prolongation of progression-free interval (PFI). Conversely, in patients who do not form TLS-like structures post-BCG, even with BCG exposure, local immune orchestration remains suboptimal, leading to higher rates of recurrence and earlier progression to muscle-invasive disease. Clinically, TLS status after TURBT+BCG will stratify patients into responders vs. non-responders, guiding escalation strategies (e.g., early re-resection, intensified or combination immunotherapies, or systemic therapy) to improve PFI. Secondary modifiers (e.g., cisplatin-based regimens or other intravesical agents) may modulate TLS formation and should be explored as adjuncts to BCG in TLS-poor tumors.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	BCG-induced local immune activation, residual tumor burden post-TURBT
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

BCG-induced local immune activation: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

residual tumor burden post-TURBT: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	cisplatin-based chemotherapy	Supporting modifier from knowledge graph

Secondary	progression to muscle-invasive bladder cancer	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: BCG exerts local immune activation in NMIBC, but heterogeneity in response suggests a need for organized adaptive responses at the mucosal site.

Prediction 2: TLSs are organized microenvironments that concentrate antigen presentation and effector lymphocytes, and their presence correlates with robust anti-tumor immunity across cancers and with favorable responses to immunotherapies.

Prediction 3: In NMIBC, residual tumor burden post-TURBT influences antigen load and the inflammatory milieu; lower residual disease may provide a more conducive setting for TLS development and effective immune priming.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- cisplatin-based chemotherapy: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- progression to muscle-invasive bladder cancer: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.